

Chapter 3 — Exercises

Multiple-Choice Questions

Unsupervised learning & preprocessing

How to Use These Slides

- **12 multiple-choice** questions on Chapter 3 concepts
- Pick **one** answer (A–D) for each question
- The **answer key** with brief explanations is at the end
- Try to answer before revealing the solutions

Topics: scaling, PCA / NMF / t-SNE, k-Means, agglomerative clustering, DBSCAN, and evaluating clusters.

Q1. The main challenge of unsupervised learning is that it:

- A. always needs a GPU
- B. is hard to evaluate because there are no labels
- C. only works on image data
- D. requires more features than samples

Q2. StandardScaler transforms each feature to have:

- A. mean 0 and variance 1
- B. values exactly in $[0, 1]$
- C. unit length per sample
- D. only integer values

Q3. A scaler should be fit on:

- A. the test data
- B. the whole dataset at once
- C. the training data only
- D. each fold's test portion

Q4. PCA orders its principal components by:

- A. alphabetical order
- B. the number of samples
- C. correlation with the labels
- D. the amount of variance they capture

Q5. t-SNE is primarily used for:

- A. visualizing high-dimensional data in 2-D
- B. transforming new data for a model
- C. supervised classification
- D. feature scaling

Q6. MinMaxScaler shifts the data into the range:

- A. mean 0, variance 1
- B. $[-1, 1]$
- C. $[0, 1]$
- D. unbounded

Q7. k-Means clustering requires you to:

- A. label some of the data first
- B. specify the number of clusters in advance
- C. scale the target variable
- D. build a dendrogram

Q8. An advantage of DBSCAN is that it:

- A. always needs the number of clusters set
- B. only finds spherical clusters
- C. cannot detect outliers
- D. finds arbitrary shapes and flags noise, without setting a cluster count

Q9. Compared to PCA, NMF produces components that are:

- A. non-negative and additive ("parts")
- B. always orthogonal
- C. ordered by variance
- D. computed from the labels

Q10. To evaluate clustering against known labels, use:

- A. `accuracy_score`
- B. `r2_score`
- C. `adjusted_rand_score` (ARI)
- D. `mean_squared_error`

Q11. Why is `accuracy_score` wrong for evaluating clusters?

- A. It is too slow to compute
- B. Cluster label numbers are arbitrary
- C. It needs probability estimates
- D. It only works for regression

Q12. Agglomerative clustering works by:

- A. randomly assigning points to k groups
- B. iteratively updating cluster centers
- C. expanding dense regions
- D. bottom-up merging of the closest clusters

Answer Key

Solutions & Explanations

Answers 1–6

Q	Ans	Why
1	B	No labels → no ground truth → hard to evaluate
2	A	<code>StandardScaler</code> → mean 0, variance 1 per feature
3	C	Fit on training only ; fitting on test = leakage
4	D	Components are ordered by variance captured
5	A	t-SNE is a visualization method (no <code>transform</code>)
6	C	<code>MinMaxScaler</code> squeezes data into [0, 1]

Answers 7–12

Q	Ans	Why
7	B	k-Means needs the number of clusters up front
8	D	DBSCAN finds any shape, flags noise , no k needed
9	A	NMF components are non-negative and additive
10	C	Use ARI / NMI with ground truth (never accuracy)
11	B	Cluster label numbers are arbitrary
12	D	Agglomerative = bottom-up merging → dendrogram

Well done!

Review Chapter 3 for any you missed